`aivreg` 1.1.2: Package Update Report

31 August 2026

1 What was merged

I merged Alex's working candidate, `aivreg_working_candidate_1_1_1_20260825.ado`, with Josh's most recent updates on branch `gmm_fix`, and called the result `aivreg_merged_draft_1_1_2_20260830.ado` (version 1.1.2). Table 1 lists the versions used throughout this note.

Table 1: Versions compared

Label	File	Role
Baseline	`aivreg.ado` (1.0.0)	Repository state before the merge
Candidate	`aivreg_working_candidate_1_1_1_20260825.ado`	Alex's reviewed candidate
Merged	`aivreg_merged_draft_1_1_2_20260830.ado`	1.1.1 + Josh's updates

The GMM weighting and clustering code is identical in the candidate and in the merged draft, so "candidate" and "merged" are interchangeable in Section 2.

2 Weighted and clustered GMM

Let w_i be the probability weight, $W = \sum_i w_i$, N the number of observations, $\hat{w}_i = w_i N/W$ the weight normalised to mean one, and ε_i the observation's moment contribution. Table 2 states what each version does.

In words: the baseline folded $\sqrt{w_i/W}$ into each score and then formed the outer product, so the weight entered the covariance linearly rather than squared; and it used the sum of weights in place of the observation count in the finite-sample factors, so multiplying every weight by a constant changed the reported standard errors. On the clustered path the weight was applied once inside the cluster sum and again as a cluster weight share when forming $\bar{g}$. The candidate applies mean-one normalised weights linearly to the moment and in squared form to the covariance, bases the finite-sample factors on N and G only, and builds the clustered $\bar{g}$ directly from this call's own cluster sums.

The effect is visible in tests `A11` and `A12` of Table 4, which are the same specification with every weight multiplied by 17. The baseline standard errors differ in the fifth significant digit (0.0267815 against 0.0267743); the merged standard errors agree to roughly twelve digits.

Table 2: Weight and cluster handling in the GMM engine

	Baseline	Candidate / merged
Weight in the point estimate	w_i/W (linear)	w_i/W (linear; unchanged)
Weight in the covariance "meat"	w_i/W — linear	$\hat{w}_i^2$ — squared
Finite-sample factor, unclustered	$W/(W-K)$	$N/(N-K)$
Finite-sample factor, clustered	$\frac{G}{G-1} \cdot \frac{W-1}{W-K}$	$\frac{G}{G-1} \cdot \frac{N-1}{N-K}$
Cluster score sum	$\sum_{i \in g} \sqrt{w_i/W}\, \varepsilon_i$	$\sum_{i \in g} \hat{w}_i\, \varepsilon_i$
Clustered $\bar{g}$	cluster weight share × cluster sums (weights counted twice)	$N^{-1} \sum_g \sum_{i \in g} \hat{w}_i \varepsilon_i$, consistent with the unclustered $\bar{g}$
Rescaling all weights by a constant	changes the standard errors and J	leaves $\hat{\beta}$, V and J unchanged
Zero or negative weights	silently dropped	error 459
`[aw=]` or `[fw=]` on `gmm`/`2sls`	accepted, treated as `pw`	error 101

3 Degrees of freedom for the Hansen J test

With k instruments (amenities and controls) and L anti-IVs, the moment system has $m = L(k+2)$ conditions and $p = k + 2L$ parameters. **The adopted rule is the conventional one,**

$$\mathrm{df} \;=\; m - p \;=\; L(k+2) - (k+2L) \;=\; k(L-1),$$

with $J = N\,\bar{g}' W_2\, \bar{g}$ evaluated at the second-step estimate and the efficient weighting matrix W_2, and p-value $= \Pr[\chi^2_{\mathrm{df}} > J]$. An earlier proposal set the degrees of freedom from numerical matrix ranks, $\mathrm{rank}(W_2) - \mathrm{rank}(X' W_2 X)$. That rule is *not* used.
Ranks are retained only as a validity gate. Before reporting J, the command checks $\mathrm{rank}(W_2) = m$ and $\mathrm{rank}(X' W_2 X) = p$. If either check fails, J is suppressed entirely, a warning is printed, and `e(J_status)` records the suppression; no generalised rank-based degrees of freedom are ever substituted. This is the conservative reading: a rank-deficient moment system has no valid χ^2 reference, so the statistic is withheld rather than reported against an adjusted distribution.
The J statistic is reported only on the efficient two-step GMM path. One-step GMM, `2sls`, and just-identified problems print an explicit note in place of a statistic, rather than a criterion value that would be read as a Hansen test.

4 Regression test results

The suite in `tests/aivreg_test_suite.do` runs 29 tests against each staged version and records the return code, the first coefficient, its standard error, and the J statistic with its degrees of freedom. Section A covers ordinary documented usage; Section B covers edge cases and the defects under review. Table 3 says what question each test answers; Table 4 reports both runs.

Table 3: What each test asks

Test	Question it answers
Section A — ordinary documented usage	
`A1`	Does the single-anti-IV ratio path work with the default Anderson–Rubin SEs?
`A2`	Does the ratio path work with asymptotic SEs?
`A3`	Does the ratio path work with bootstrap SEs (seeded)?
`A4`	Do factor-variable controls with `if` work on the ratio path?
`A5`	Do fixed effects work on the ratio path?
`A6`	Does a multi-anti-IV call switch to GMM automatically?
`A7`	Does two-step GMM run with multiple anti-IVs?
`A8`	Does one-step GMM run?
`A9`	Does 2SLS run?
`A10`	Does clustering (on `fips`) work under GMM?
`A11`	Do probability weights work under GMM?
`A12`	Same call as `A11` with all weights $\times 17$: are results scale invariant?
`A13`	Does the ratio path still accept its documented `[aw=]` syntax?
Section B — edge cases and known defects	
`B1`	Is an `[in]` range honoured on the ratio path?
`B2`	Is an `[in]` range honoured on the GMM path?
`B3`	Are `if` and `[in]` combined honoured on the GMM path?
`B4`	Is an explicit `[aw=]` rejected loudly on the GMM path?
`B5`	Is an explicit `[fw=]` rejected loudly on the GMM path?
`B6`	Are zero or negative weights refused loudly?
`B7`	Are observations with missing weights dropped from the sample?
`B8`	Does fewer than two clusters produce a clean error?
`B9`	Do 32-character variable names work on the ratio path?
`B10`	Do 32-character variable names work on the GMM path?
`B11`	Does a failed call avoid leaving the previous call's results in `e()`?
`B12`	Do fixed effects, clustering and weights work together?
`B13`	Is J suppressed on a just-identified problem (single anti-IV via `gmm`)?
`B14`	Does a collinear anti-IV fail rather than silently collapse the moment system?
`B15`	Does a mistyped estimator (`gmmm`) error rather than run?
`B16`	Do non-consecutive integer fixed-effect levels work?

Table 4: Baseline versus merged candidate, 29 regression tests

Test	Return code		Coefficient		Std. error		J (df)	
	Base	Merg	Base	Merg	Base	Merg	Base	Merg
A1_ratio_default	0	0	-1.1451	-1.1451	0.1011	0.1011	–	–
A2_ratio_asymp	0	0	-1.1451	-1.1451	0.1091	0.1091	–	–
A3_ratio_boot	0	0	-1.1451	-1.1451	0.1154	0.1154	–	–
A4_ratio_controls_if	0	0	-0.5250	-0.5250	0.0204	0.0204	–	–
A5_ratio_fe	0	0	-0.5250	-0.5250	0.0204	0.0204	–	–
A6_multiamenity_gmm	0	0	-0.5355	-0.5355	0.0213	0.0213	–	–
A7_gmm_twostep	0	0	-0.5369	-0.5369	0.0275	0.0275	1.86×10^{-5} (–)	0.3987 (2)
A8_gmm_onestep	0	0	-0.4237	-0.4237	0.1856	0.1856	0.0074 (–)	–
A9_2sls	0	0	-0.5385	-0.5385	0.0279	0.0279	0.9993 (–)	–
A10_gmm_cluster	0	0	-0.5222	-0.5222	0.0590	0.0590	6.70×10^{-7} (–)	8.7252 (2)
A11_gmm_pweight_bare	0	0	-0.5320	-0.5319	0.0268	0.0278	2.16×10^{-5} (–)	0.4773 (2)
A12_gmm_pweight_x17	0	0	-0.5320	-0.5319	0.0268	0.0278	2.16×10^{-5} (–)	0.4773 (2)
A13_ratio_aw_bracket	0	0	-0.8091	-0.8091	0.0345	0.0345	–	–
B1_ratio_in	0	0	-2.9689	-2.9689	0.1485	0.1485	–	–
B2_gmm_in	0	0	-1.6825	-1.6825	0.1161	0.1161	0.0082 (–)	339.3547 (2)
B3_gmm_if_in	0	0	-1.6289	-1.6289	0.1163	0.1163	0.0008 (–)	9.8732 (2)
B4_gmm_aw_bracket	101	101	-1.6289	–	0.1163	–	0.0008 (–)	–
B5_gmm_fw_bracket	101	101	-1.6289	–	0.1163	–	0.0008 (–)	–
B6_gmm_negzero_weight	0	459	-0.5047	–	0.0267	–	2.54×10^{-5} (–)	–
B7_gmm_missing_weight	0	0	-0.5527	-0.5527	0.0295	0.0306	2.30×10^{-6} (–)	0.0270 (2)
B8_gmm_two_clusters	430	430	0.0000	0.0000	0.0000	0.0000	0.2499 (–)	–
B9_ratio_longnames	111	0	3.0944	-0.8396	0.1101	0.0392	–	–
B10_gmm_longnames	198	0	-0.8386	-0.8394	0.0687	0.0460	9.01×10^{-8} (–)	0.0002 (1)
B11_state_after_error	0	459	-0.5047	–	0.0267	–	2.54×10^{-5} (–)	–
B12_gmm_fe_clus_w	0	0	-0.5279	-0.5279	0.0579	0.0586	1.09×10^{-8} (–)	2.66×10^{-6} (1)
B13_gmm_justidentified	0	0	-0.5293	-0.5293	0.0279	0.0279	–	–
B14_gmm_collinear_aiv	430	430	0.0356	0.0356	0.0053	0.0053	0.9994 (–)	–
B15_bad_estimator	0	0	-0.8386	-0.8386	0.0687	0.0687	9.01×10^{-8} (–)	–
B16_fe_nonconsecutive	0	0	-0.5250	-0.5250	0.0204	0.0204	–	–

Notes. "–" means the quantity was not returned. Baseline rows with a non-zero return code still report a coefficient because the baseline left the previous call's results in `e()`; the merged draft clears them. Return code 101 is the refusal of `[aw=]` or `[fw=]` on the GMM path, 459 the refusal of non-positive weights, 430 the singular anti-IV system, and 111 and 198 the long-name failures the merge repairs.

Reading the table

- **Hansen J.** The J calculation is now correct. The baseline returned a degenerate near-zero J with no degrees of freedom on every over-identified call; the merged draft reports a proper statistic with its degrees of freedom (`A7`, `A10`) and suppresses J where the test is not defined.
- **Loud failures.** `B4`, `B5`, `B6` and `B11` now exit with an error and leave no results, where the baseline either ran or reported stale values.
- **Long names.** `B9` and `B10` fail under the baseline (return codes 111 and 198) and succeed under the merge.